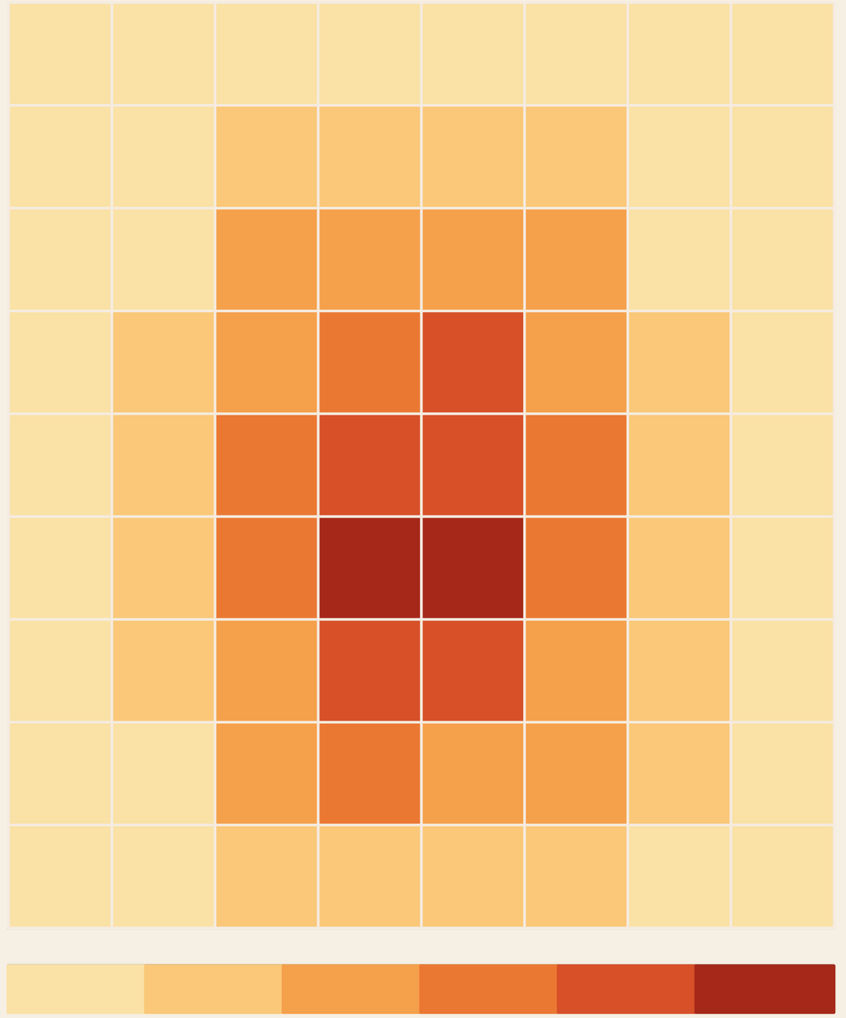
Phoenix HeatShield

Explainable Heat Exposure and Cooling Priority Mapping

The Problem

- Extreme urban heat does not affect every community in Phoenix equally.
- Vegetation cover, built surfaces, and access to resources shape how much heat a community is exposed to and how well it can cope.
- City-wide averages hide which of the 378 census tracts need attention first.



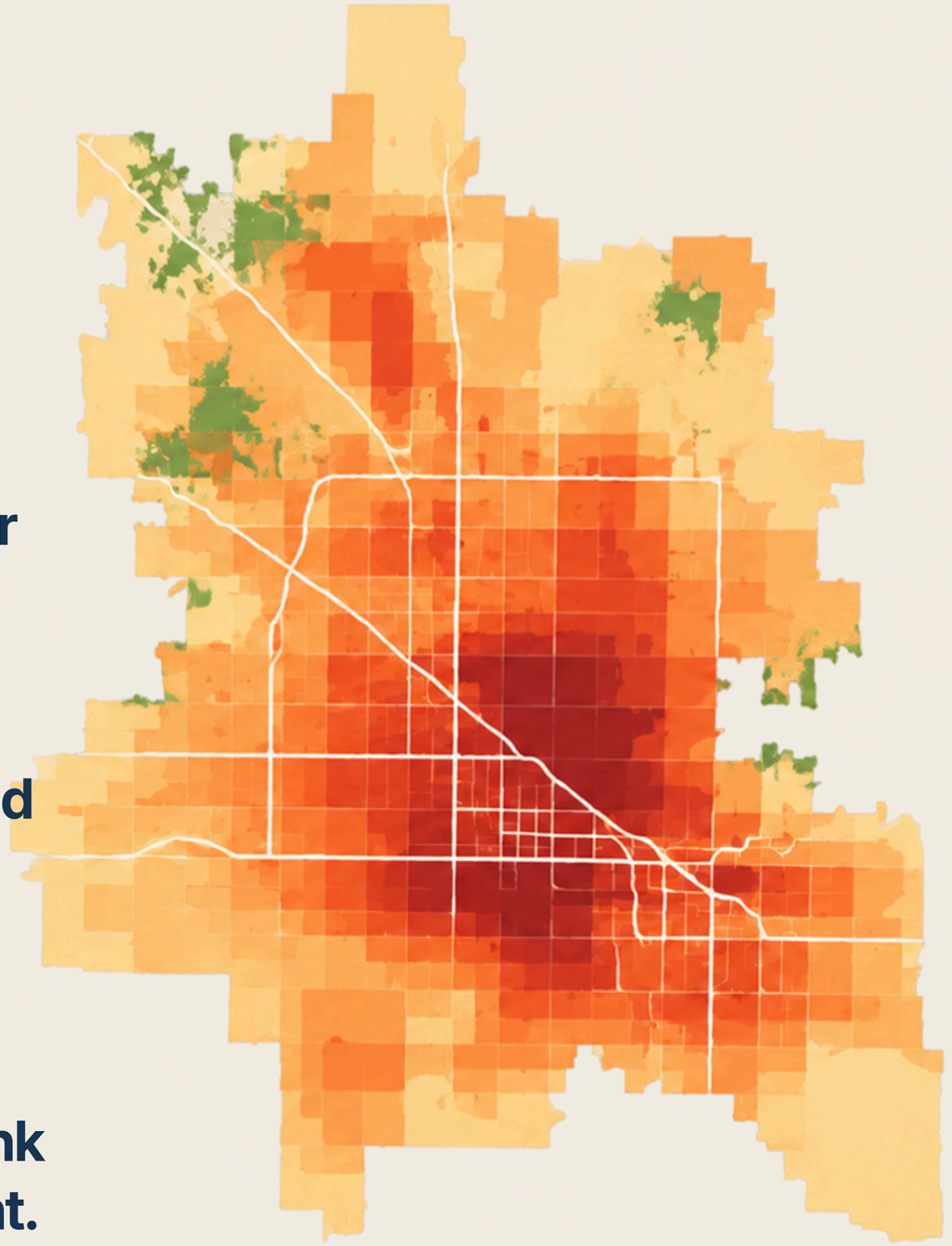
Cooler — Hotter

Surface temperature

(illustrative)

Our Solution

- We will build a web-based system for the 378 census tracts in Phoenix, Arizona.
- We will combine satellite-derived land surface temperature, environmental indicators, and socioeconomic data.
- We will model spatial variation in daytime surface temperature and rank tracts for cooling-related assessment.



Surface temperature

Vegetation cover

Built surfaces

Access to resources

Illustrative concept

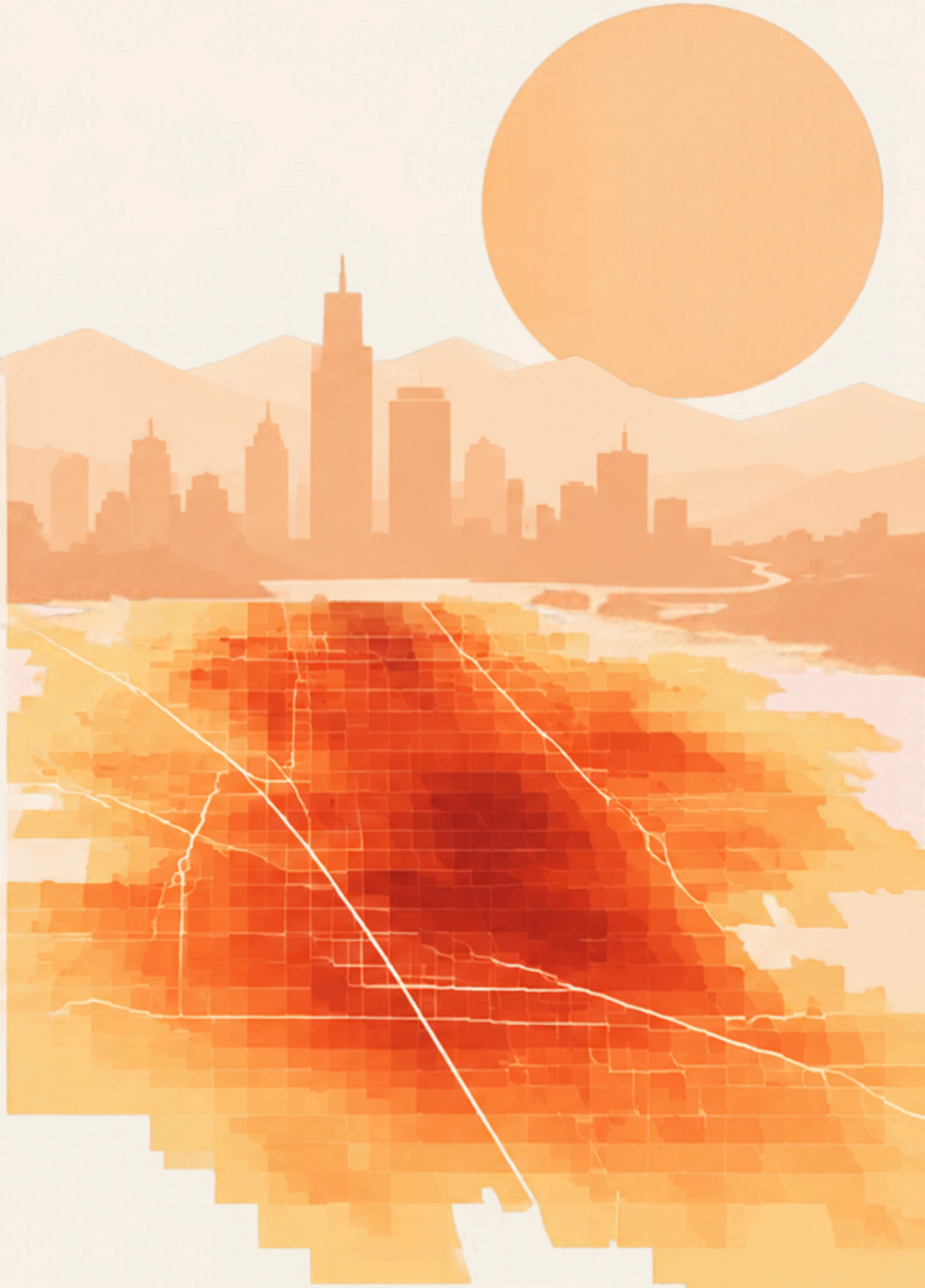
Alignment with global goals

SDG 11 and 13

Evidence-based urban adaptation and local focus on regions affected by extreme heat.

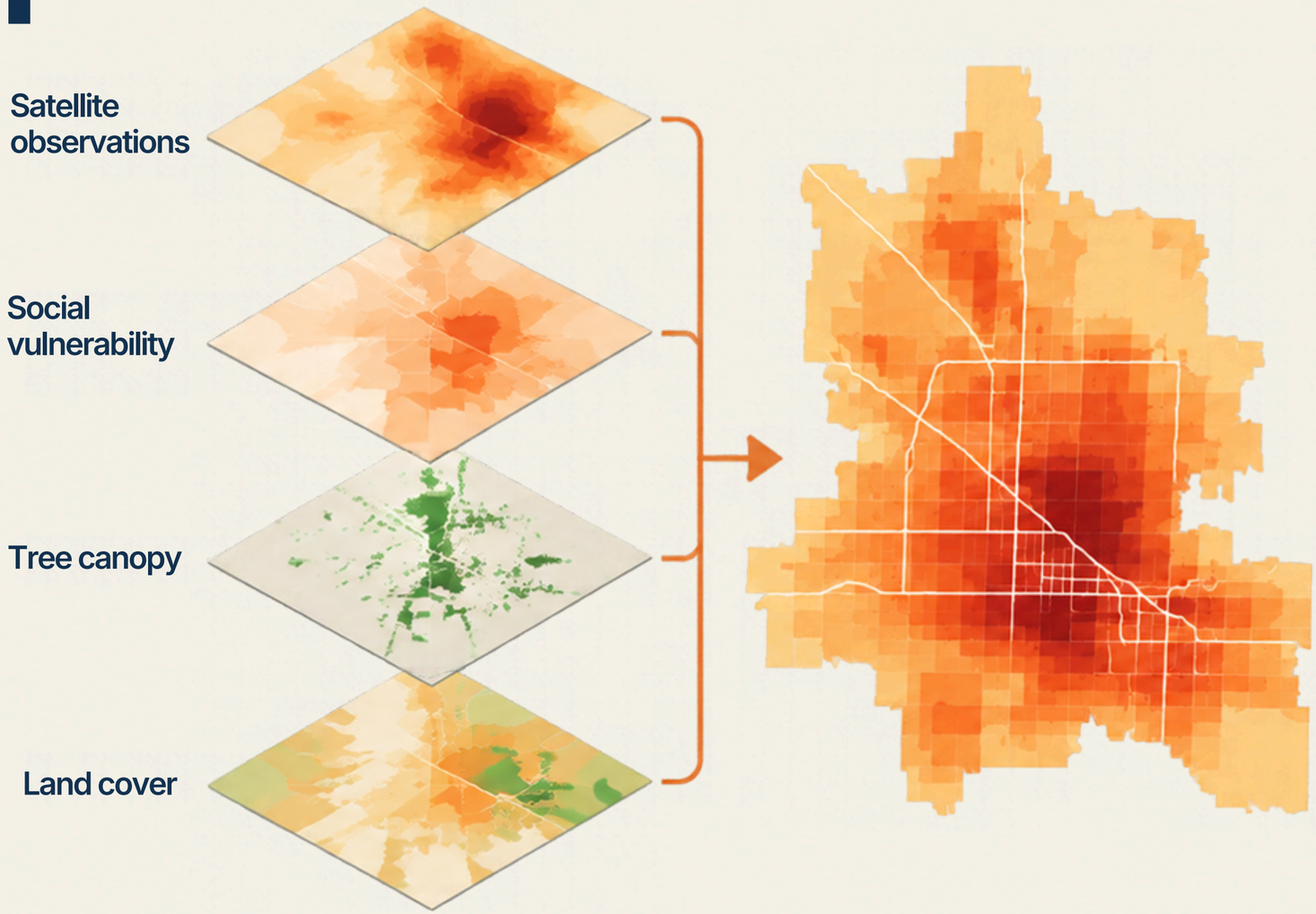
SDG 10

Income and access to transportation highlight disparities in communities' heat-coping capacity.



Learning from the literature

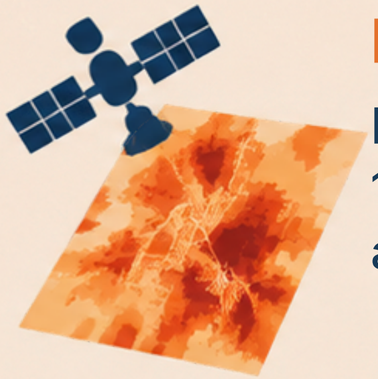
- Bautista et al. (2022) combined satellite observations, social vulnerability, and tree-canopy data to prioritize tree planting in Phoenix. We build on their spatial integration and prioritization approach [1].
- Li et al. (2016) examined how land-cover composition, configuration, and socioeconomic conditions relate to surface temperature in Phoenix. We use their findings to select our predictors [2].



[1] Bautista et al. (2022). NASA DEVELOP technical report, Phoenix, AZ.
[2] Li et al. (2016). Remote Sensing of Environment, 174, 233-243. doi:10.1016/j.rse.2015.12.022

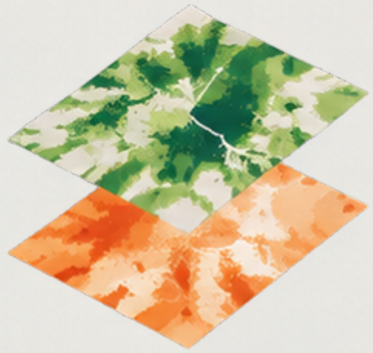
Illustrative concept

Our data and scope



NASA MODIS:

MOD11A2.061
1 km daytime LST
and quality flags.

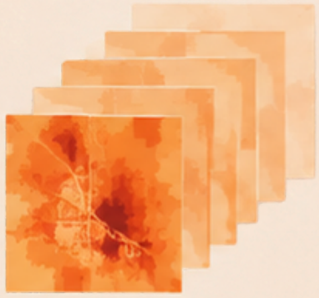


CAP LTER:

30 m NDVI
and ENDISI
rasters.

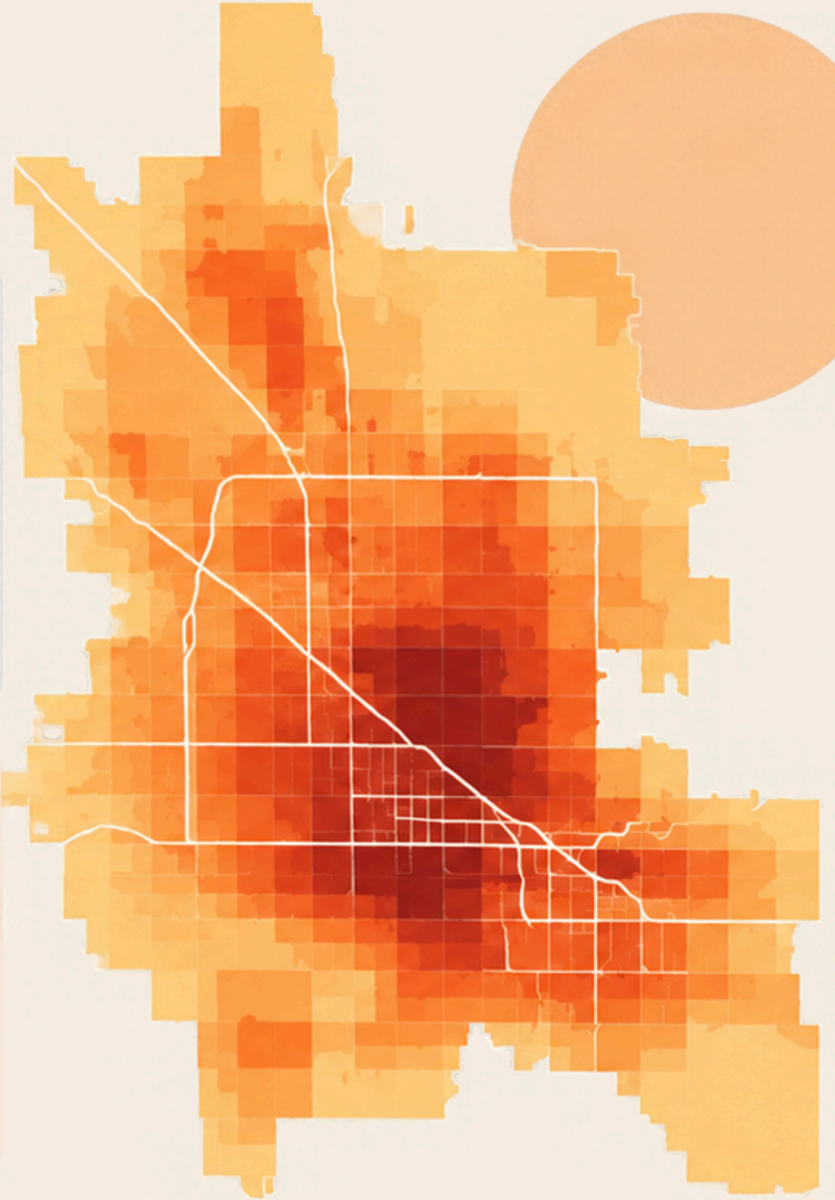
Phoenix GIS:

Tree canopy,
income, and
vehicle access



378 census tracts

Nine eight-day LST
composites starting
May 25 to
July 28, 2023



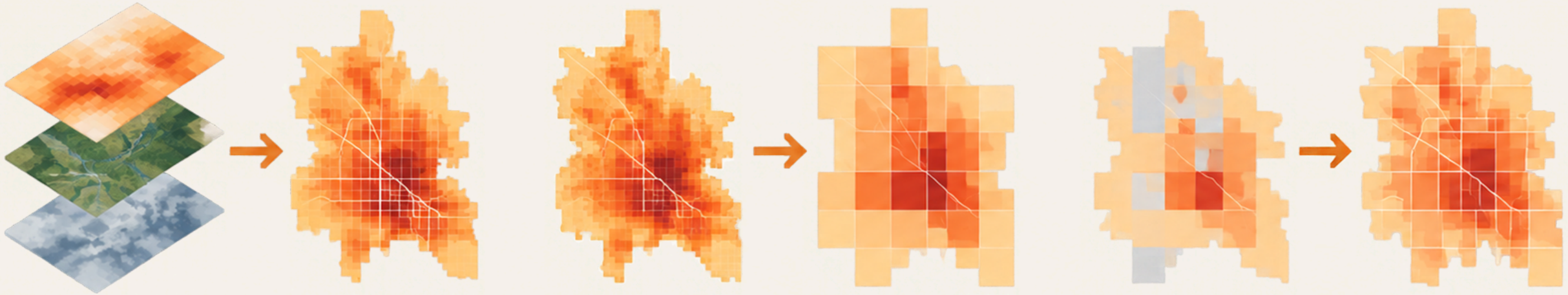
Illustrative concept

Preprocessing Workflow

1. Align coordinates and select seasonal bands.

2. Aggregate data to tracts, accounting for quality flags.

3. Handle missing values and verify tract-level joins.

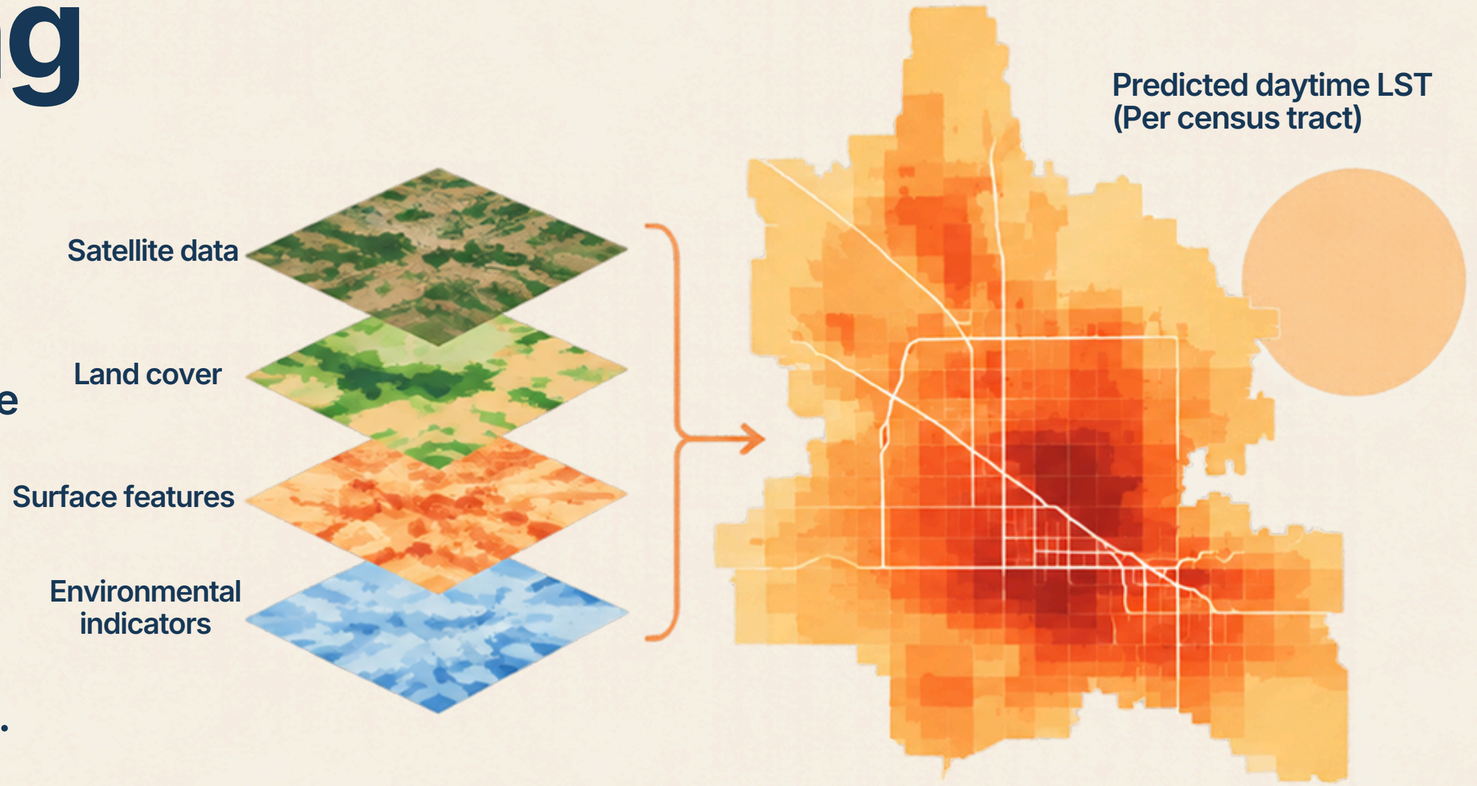


Illustrative concept

Machine Learning Approach

Random Forest Regression

- We will use Random Forest Regression to estimate tract-level daytime LST from environmental predictors.
- We will compare it against a linear regression baseline.
- We will use spatial cross-validation with MAE and RMSE to measure performance on held-out areas.



Environmental predictors
NDVI / ENDISI
Tree canopy

Random Forest Regression
Daytime LST predictions

Linear regression baseline
Daytime LST predictions

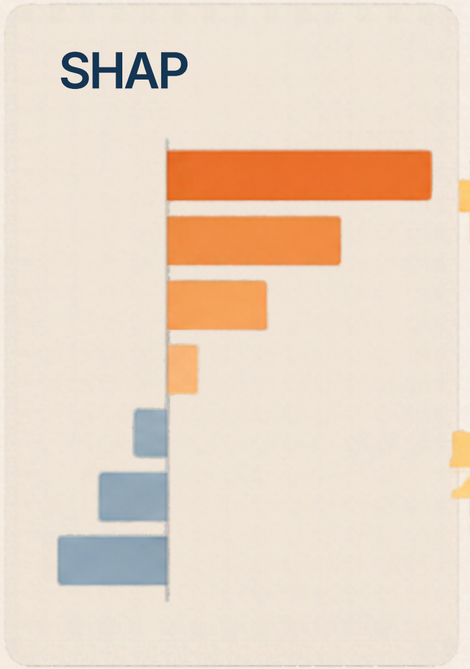
Spatial cross-validation
Same held-out areas
MAE / RMSE

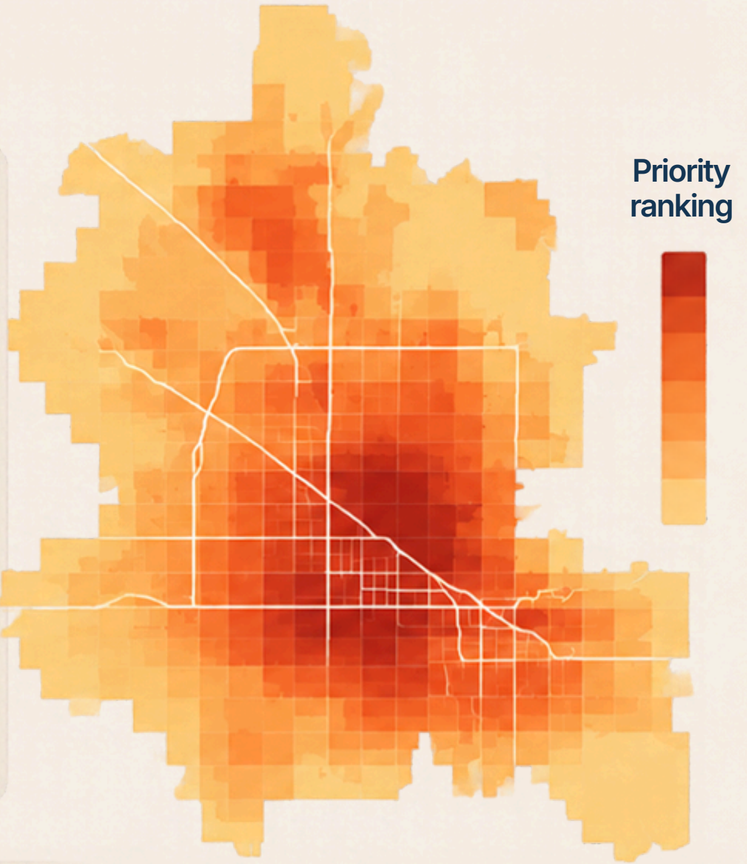
Compare models
Observed daytime LST as the reference

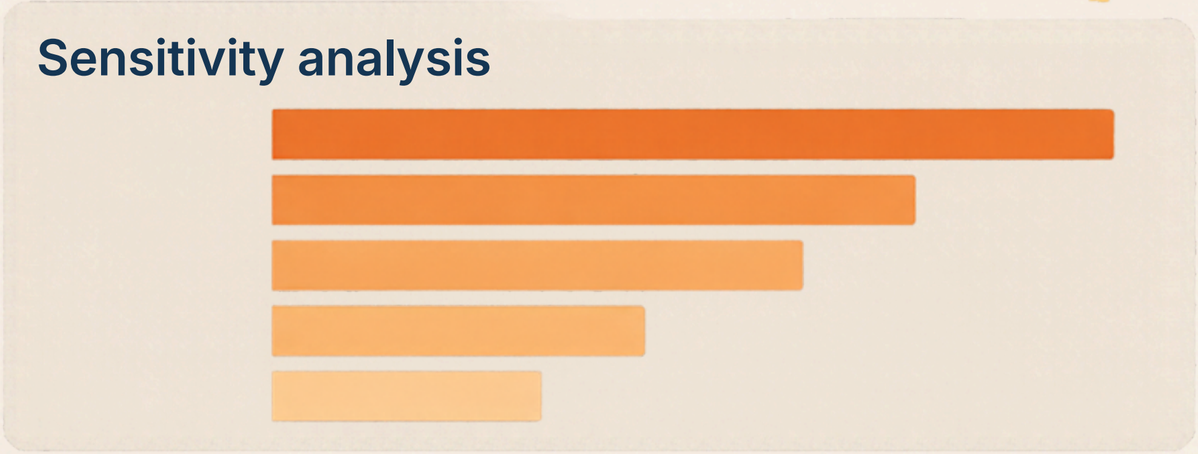
Illustrative concept

Explanation and prioritization

- SHAP explains which indicators are associated with each tract's predicted temperature
- Priority ranking: We will combine thermal exposure with income and vehicle access into a transparent ranking.
- Sensitivity analysis: We will test how changing indicator weights affects the ranking.

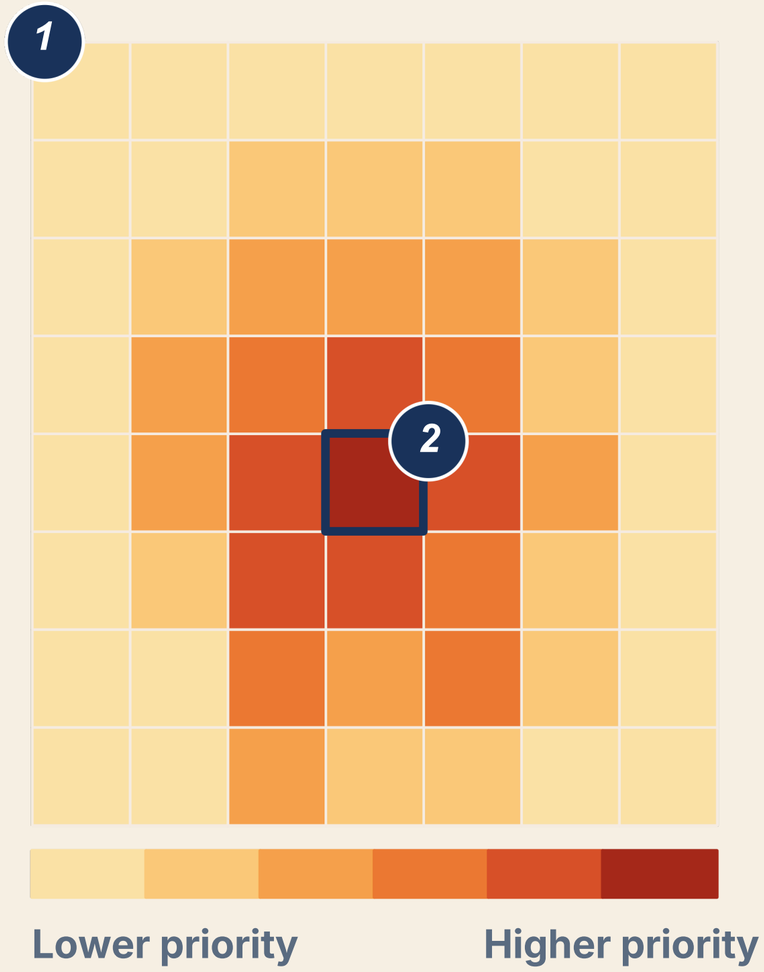


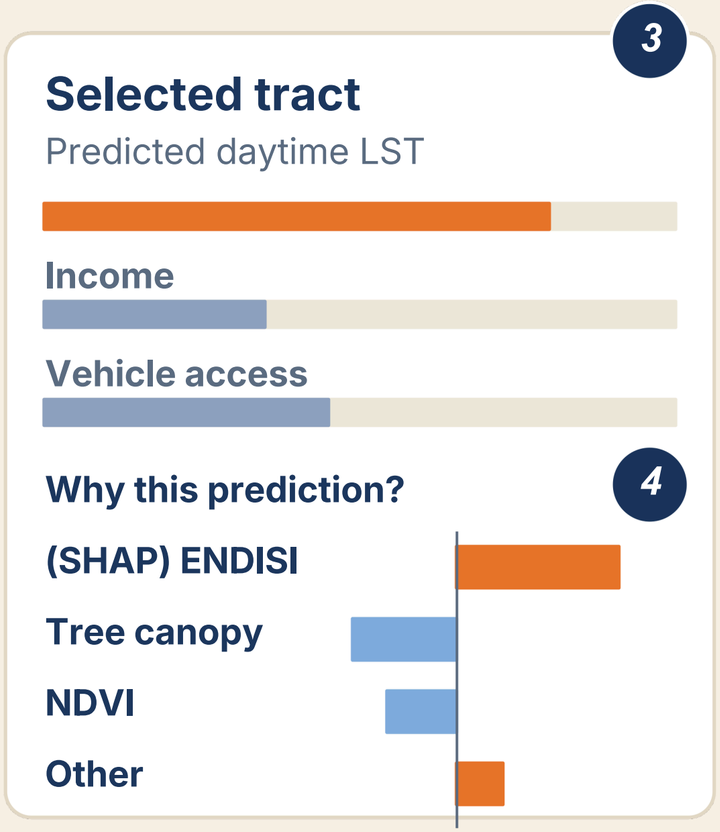




Illustrative concept

What our output will show



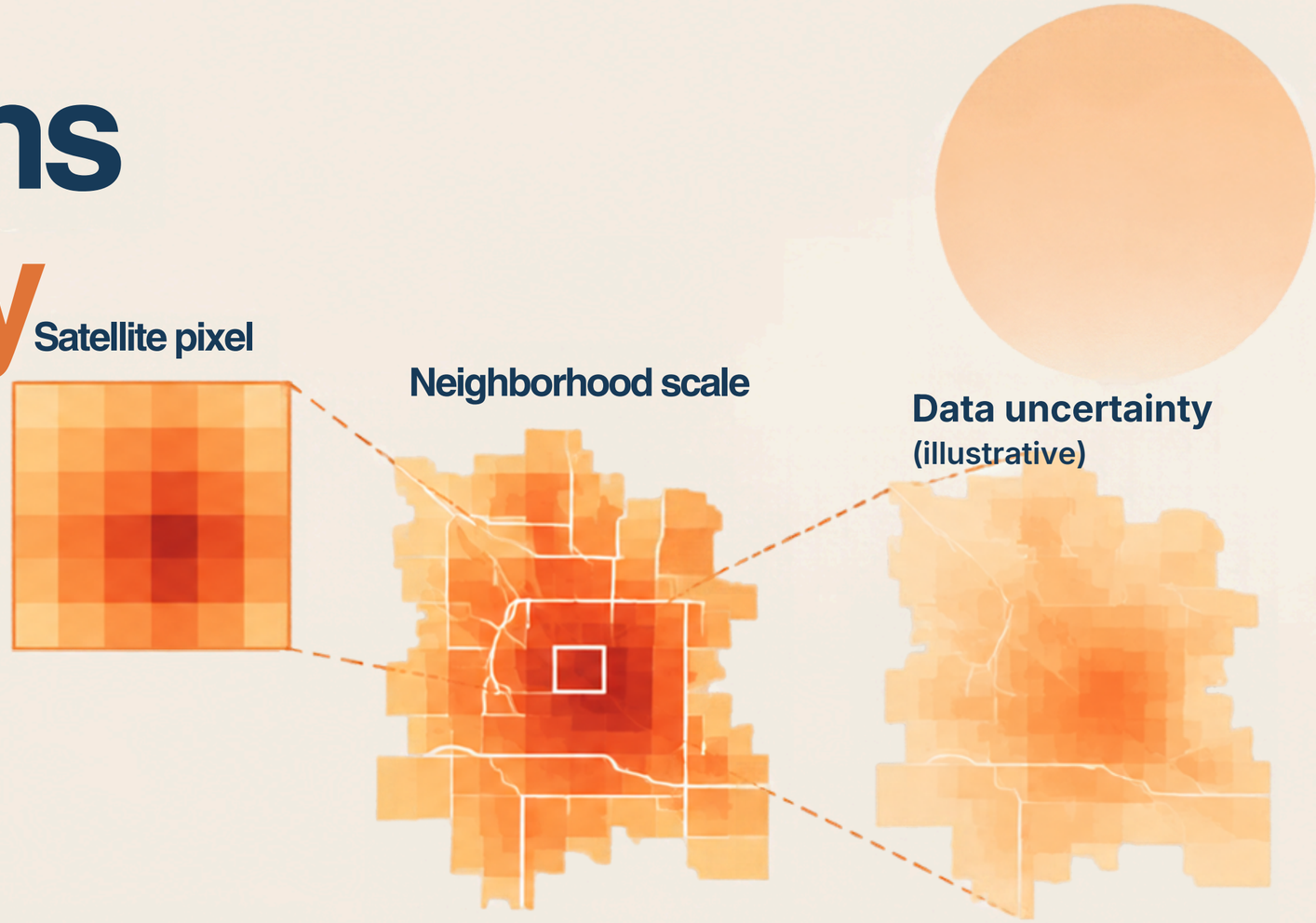


1. **Map:** Each cell is a census tract. Darker color means higher cooling priority.
2. **Selected tract:** Clicking a tract opens its detail panel.
3. **Tract details:** Predicted daytime LST plus income and vehicle access.
4. **Reasons:** SHAP bars show which indicators push the predicted temperature up (orange) or down (blue).

Illustrative concept

Model limitations and uncertainty

- Land surface temperature is not the same measurement as air temperature.
- MODIS resolution may limit what we can see at the scale of small neighborhoods.
- Quality filtering and missing observations may affect LST estimates and coverage.
- We will interpret our results within the limits of the measurement and data coverage.



Illustrative concept

Make heat risk visible.

Explainable maps to help
prioritize cooling assessment in Phoenix.